Patent phrase matching

Utkrisht Suman | A24ARIU0001 | Measured experiment report

DRAFT: scores verified through Kaggle API; required Submissions-tab screenshot is missing. Course-platform upload remains pending.

1. Validation design and baselines

The model predicts a continuous similarity score for an anchor-target pair in CPC context. The supplied baseline uses DeBERTa-v3-small with a regression head and mean squared error. The seed-42 GroupShuffleSplit holds out 20% of anchor groups. Random row splitting would allow the same anchor to occur in both sets and could overstate generalization.

Run / changed factor	Validation r	Training min	Peak GB
section_baseline none (starter)	0.8202	3.43	2.73
minilm_no_training bi-encoder; different pretrained checkpoint; no task training	0.5814	0.00	-
context_none context only: none	0.8167	3.33	2.66
context_code context only: code	0.8192	3.31	2.71

The pretrained MiniLM bi-encoder, with no task training, obtained r=0.5814; the fine-tuned cross-encoder obtained r=0.8202 (difference +0.2387). MiniLM encodes each phrase separately and compares normalized mean-pooled embeddings. The cross-encoder attends jointly across both phrases. The comparison changes the checkpoint and training regime together; it cannot isolate the effect of fine-tuning alone.

Training: maximum length 64, batch 32, three epochs, learning rate 2e-5, warmup ratio 0.1, weight decay 0.01. Timings cover Trainer.train(), including epoch evaluation; MiniLM has no training time. Peak memory is allocated CUDA memory, excluding other processes and some reserved memory.

2. Controlled context experiments

context_none: only the context representation changes; delta r=-0.0035. The difference is at most 0.005 and may be seed noise.

context_code: only the context representation changes; delta r=-0.0010. The difference is at most 0.005 and may be seed noise.

Selected for the final run: section_baseline using validation r=0.8202. All variants use the same validation IDs and initial checkpoint. No leaderboard score was used for selection.

3. Section and score analysis

Section	N	Pearson r
A	943	0.7858
B	1926	0.8938
C	1177	0.7692
D	199	0.7892
E	373	0.8088
F	968	0.8475
G	1040	0.7062
H	933	0.8050

The lowest measured section is G (n=1040). Five of the ten largest errors are in G. The smallest group, D (199 pairs), is not the lowest. Technical relations under coarse context are a hypothesis to investigate, not an established cause.

True score	N	Mean prediction
0.00	1522	0.119
0.25	2465	0.305
0.50	2475	0.492
0.75	832	0.661
1.00	265	1.025

Zero labels move inward and 0.75 is underpredicted, but the mean at 1 overshoots: compression is not uniform. The dominant 0.25/0.5 labels and squared-error regression can contribute to central tendency, but do not explain every group. These are raw predictions; only submission predictions are clipped to [0,1]. Pearson alone does not establish calibration.

4. Largest validation errors

These ten pairs have the largest absolute errors for the validation-selected model. Categories and explanations below come from the reviewed error CSV; disagreement alone does not establish that an expert label is wrong.

Anchor / target / context	True / predicted	Reviewed failure and evidence
punch face / face punch [B05]	0.00 / 0.917	Relation-sensitive lexical match: Reversing 'punch face' to 'face punch' can change a tool surface into an action. The high score is consistent with lexical confusion; its cause is not established. Label debatable: no clear issue
boom hydraulic cylinder / boom a hydraulic cylinder [E02]	0.00 / 0.830	Relation-sensitive lexical match: The target inserts 'a' into almost the same content words. The model scores 0.830 despite the zero label. Label debatable: possibly; expert context needed
imaging axis / imaging of axis [G02]	0.00 / 0.820	Relation-sensitive lexical match: An imaging axis names an axis; imaging of an axis names a process applied to it. The model largely treats them alike. Label debatable: no clear issue
imaging axis / image of axis [G02]	0.00 / 0.766	Relation-sensitive lexical match: An imaging axis and an image of an axis need not denote the same thing; adding 'of' changes the relation. Label debatable: no clear issue
end pins / wood pin [G10]	0.75 / -0.008	Technical or part-whole relation: The anchor names a pin component while the target adds a material/type. The model misses the annotated strong relation. Label debatable: possibly; expert context needed
breech mechanism / breech loading naval gun [B29]	0.75 / 0.005	Technical or part-whole relation: A breech mechanism is a component; the target names a breech-loading gun. Related granularity is missed, with prediction 0.005. Label debatable: no clear issue
average pore size / average cell size [B24]	0.75 / 0.040	Technical or part-whole relation: Both phrases measure structural size, but 'pore' versus 'cell' needs a technical sense. Prediction 0.040 misses the 0.75 label. Label debatable: possibly; expert context needed
return structure / return to structure [D01]	0.00 / 0.698	Relation-sensitive lexical match: 'Return structure' can name a structure; 'return to structure' expresses a direction/action. The high prediction misses this possible relation change. Label debatable: no clear issue
whose identity / customer id [G06]	0.75 / 0.066	Functional paraphrase missed: The phrases concern identity versus a customer identifier, but the wording and specificity differ. The model misses annotated relatedness. Label debatable: possibly; expert context needed
running tally / credit total [G08]	0.75 / 0.099	Functional paraphrase missed: A running tally and a credit total both concern accumulated quantities; little literal overlap accompanies a low prediction. Label debatable: no clear issue

The complete unabridged notes and validation predictions accompany this report. Error categories are descriptive observations from these cases; the ten largest errors are not a representative sample of the whole validation set.

5. Late submissions and score comparison

Version / status	Public r	Private r
1 / complete	0.7922	0.8084

Version 1: public-minus-validation -0.0280; private-minus-validation -0.0118.

The final saved run independently reproduces validation r=0.820194478; its Trainer-reported runtime is 214.4694 seconds. For the recorded submission, both hidden scores are lower and the private result is closer to validation.

Validation, public and private partitions can differ in anchor mix, CPC sections and size. Selecting settings using validation also introduces selection optimism. The final notebook trains only the validation-selected variant, loads attached local weights with internet off, and writes id,score in submission.csv. The same grouped training budget is retained for reproducibility.

Every submission is listed above, including failures. Version URLs and screenshot evidence:

1: https://www.kaggle.com/code/utkrishtsuman/a24ariu0001-patent-final?scriptVersionId=367835971
Screenshot: MISSING
Notes: Submission 58176236; all attempts listed (one). Scores verified using official Kaggle API. Submissions-tab screenshot missing.

References and attribution

Course patent_phrase_starter.ipynb, download_weights.ipynb and project_brief.md, bk-anupam/generative_ai_shared (adapted starter code).

Kaggle: U.S. Patent Phrase to Phrase Matching, competition data and rules. https://www.kaggle.com/competitions/us-patent-phrase-to-phrase-matching

He et al. (2021). DeBERTaV3. arXiv:2111.09543. Model: microsoft/deberta-v3-small (MIT).

Reimers & Gurevych (2019). Sentence-BERT. arXiv:1908.10084. Model: sentence-transformers/all-MiniLM-L6-v2 (Apache-2.0).

Aslanyan & Wetherbee (2022). Patents Phrase to Phrase Semantic Matching Dataset. arXiv:2208.01171. Only competition train/test files used; no released test labels.